



INTE 22283 – Mobile Application Development

Course Outline

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Overview

- Learning Outcome
- Course Content
- Recommended Reading
- Evaluation



Learning Outcomes

On completion of this course unit, the student should be able to:

- Describe the characteristics of mobile networks & communication.
- Choose a suitable mobile development platform based on requirements.
- Design a proper flow of user interfaces for a mobile application.
- Develop and test mobile applications.
- Enforce proper security mechanisms to mobile apps.
- Work with mobile iot devices.



Course Content

- Cellular networks and mobile networking
- Mobility of computers
- Mobile operating systems
- Mobile application development platforms
- mobile architecture
- Integrated Development Environments for mobile application development
- Introduction to mobile UI design patterns
- SOAP and REST services integration
- Introduction to location services
- Mobile application monetization
- Mobile application testing
- Use of smartphone sensors
- Database Connectivity
- Hybrid mobile apps
- Mobile app frameworks
- Mobile security
- IoT with mobile computing



Recommended Reading

- Joseph Annuzzi Jr., Lauren Darcey, Shane Conder, Introduction to Android Application Development: Android Essentials, 5th Edition, Addison Wesley, 2015
- Jim Doherty, Wireless and Mobile Device Security, Jones & Bartlett Learning, LLC, 2016
- Bill Phillips, Chris Stewart & Kristin Marsicano, Android Programming: The Big Nerd Ranch Guide, 3rd Edition, Big Nerd Ranch Guides, 2017
- Dawn Griffiths, David Griffiths, Head First Android Development, 2nd Edition, O'Reilly, 2017
- Madona S. Wambua, Modern Android 13 Development Cookbook, Packet Publishing, 2023
- Material provided in eKel.



Evaluation

- Individual Assignment 10%
 - Group Project 30%
 - Final Examination 60%
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- **In case of absence for a group presentation, a penalty of zero marks will be given.
 - **It is mandatory to get at least 40% for the Final Exam to pass the module.

